

Linear regression and causality

ECON726

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Potential Outcomes

Conditional Independence Assumption

Regression and causality

Regression gives the best linear approximation to the CEF, in a MSE sense

- ▶ But when does a regression represent something causal?

A regression is **causal** when the CEF it approximates is causal

- ▶ Helpful to think of a CEF being causal by using the potential outcomes framework
- ▶ CEF is causal when it describes differences in average potential outcomes for a fixed reference population

Review - Potential Outcomes framework

Consider the relationship between future earnings Y_i and whether i attends college, denoted by D_i

$D_i \in \{0, 1\}$	Binary treatment variable for individual i
Y_i	Outcome Measure of earnings for individual i
$Y_i(1)$	Treatment potential outcome Earnings if individual i goes to college ($D_i = 1$) Alternative notation: $Y_i(1)$, Y_i^1 , y_{i1}
$Y_i(0)$	Control potential outcome Earnings if individual i does not go to college ($D_i = 0$) Alternative notation: $Y_i(0)$, Y_i^0 , y_{i0}

- ▶ $Y_i(0)$ is earnings of an individual had they not gone to college, *irrespective of whether they actually went*
- ▶ Causal effect college attendance for individual i is the difference $Y_i(1) - Y_i(0)$

Observed outcome Y_i can be written as:

$$\begin{aligned}
 Y_i &= \begin{cases} Y_i(1) & \text{if } D_i = 1 \\ Y_i(0) & \text{if } D_i = 0 \end{cases} \\
 &= Y_i(0) + (Y_i(1) - Y_i(0))D_i
 \end{aligned}$$

We can see $Y_i(0)$ or $Y_i(1)$ in the data, but never both

Review - Fundamental problem of causal inference

Problem:

- ▶ If an individual is treated ($D_i = 1$), we never observe the outcome in which they are not treated ($Y_i(0)$)
- ▶ If an individual is not treated ($D_i = 0$), we never observe the outcome in which they are treated ($Y_i(1)$)

The causal effect $Y(1) - Y(0)$ is unidentifiable

Group	$Y(1)$	$Y(0)$
$D = 1$	Observable as Y	Unobservable counterfactual
$D = 0$	Unobservable counterfactual	Observable as Y

Y_1	Y_0
y_{11}	y_{10}
y_{21}	y_{20}
y_{31}	y_{30}
$\vdots$	$\vdots$
y_{n1}	y_{n0}

Stable Unit Treatment Value Assumption (SUTVA)

To link potential outcomes to observed data, we need two assumptions; together, these are known as **SUTVA**

1. Consistency: we observe $Y_i = Y_i(D_i)$

- ▶ Equivalently, $Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$
- ▶ Rules out hidden versions of treatment: $D_i = 1$ must mean the same thing for everyone
- ▶ e.g., if “college” includes both a 2-year and 4-year degree, $Y_i(1)$ is not well-defined

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2. No interference: i 's potential outcomes depend only on i 's own treatment, not on anyone else's

- ▶ Built into the notation $Y_i(d)$, which is indexed only by i 's own treatment status
- ▶ Rules out spillovers, externalities, and general equilibrium effects

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When might no interference fail?

- ▶ Vaccines: if many people are vaccinated, herd immunity protects the unvaccinated too
- ▶ College: if many people get a degree, the college wage premium may fall

Types of comparisons

Causal effect of interest is often **Average Treatment Effect**:

$$E[Y(1) - Y(0)] = E[Y(1)] - E[Y(0)]$$

or **Average Treatment Effect on the Treated**:

$$E[Y(1) - Y(0)|D = 1] = E[Y(1)|D = 1] - E[Y(0)|D = 0]$$

We often discuss what is directly observable in data as the **naive comparison**. In the population this is:

$$E[Y|D = 1] - E[Y|D = 0]$$

This is also the **average predictive effect** of treatment status on outcome.

Naïve comparison: $E[Y|D = 1] - E[Y|D = 0]$

Let's consider the averages of $Y_i(1)$, $Y_i(0)$, and Y_i :

$$\begin{aligned} E[Y|D = 1] &= E[Y(0) + (Y(1) - Y(0))D|D = 1] && \because Y = Y(0) + (Y(1) - Y(0))D \\ &= E[Y(1)|D = 1] \\ E[Y|D = 0] &= E[Y(0)|D = 0] \end{aligned}$$

We can re-write the average predictive effect (observed difference) as:

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Naïve comparison (cont.)

Observed difference in earnings equals the average treatment effect (ATE) on the treated plus selection bias:

$$\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0] = \underbrace{\mathbf{E}[Y(1) - Y(0)|D = 1]}_{\text{ATE on treated}} + \underbrace{\mathbf{E}[Y(0)|D = 1] - \mathbf{E}[Y(0)|D = 0]}_{\text{Selection bias}}$$

ATE on treated is the average causal effect of college attendance on those who went to college:

$$\mathbf{E}[Y(1)|D = 1] - \mathbf{E}[Y(0)|D = 1] = \mathbf{E}[Y(1) - Y(0)|D = 1]$$

Selection bias:

$$\mathbf{E}[Y(0)|D = 1] - \mathbf{E}[Y(0)|D = 0]$$

What do you think selection bias looks like in this example?

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What do you think selection bias looks like in this example?

- ▶ The difference in $Y_i(0)$ between those who went to college and those who didn't
- ▶ Consider someone who went to college. Now assume that person didn't go. Their expected earnings may be higher than those of someone who didn't go to college at all
 - ⇒ Those who didn't go to college, on average, have lower values of $Y_i(0)$
 - ⇒ Selection bias is positive

Example: Marijuana and longevity

Does smoking marijuana (D_i) affect longevity (Y_i)?

- ▶ Suppose marijuana has **no causal effect** on longevity: $Y_i(1) = Y_i(0)$ for everyone
- ⇒ ATE on treated = $E[Y(1) - Y(0)|D = 1] = 0$

But smoking is not randomly assigned: people choose whether to smoke

- ▶ Suppose smokers are more likely to make other poor health choices (e.g., drinking alcohol) that shorten life expectancy

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Then smokers would have shorter lives *even if they had not smoked*:

$$\underbrace{E[Y|D = 1] - E[Y|D = 0]}_{\text{Naïve comparison} < 0} = \underbrace{0}_{\text{ATE on treated}} + \underbrace{E[Y(0)|D = 1] - E[Y(0)|D = 0]}_{\text{Selection bias} < 0}$$

- ▶ Data show a negative relationship between smoking and longevity, even though the causal effect is zero
- ▶ Selection bias is negative here; in the college example it was positive

Random assignment and selection bias

Random assignment assumption: treatment status is statistically independent of each potential outcome, and assignment is nontrivial:

$$D \perp Y(d) \text{ for } d \in \{0, 1\}, \quad \text{and} \quad 0 < P(D = 1) < 1$$

- ▶ Treatment assignment mechanism is purely random; knowing D tells us nothing about $Y(0)$ or $Y(1)$
- ▶ **Positivity** ($0 < P(D = 1) < 1$) ensures there are units in both treatment and control

Random assignment ($Y(0), Y(1) \perp D$) will solve the selection problem

- ▶ Causes selection bias to equal zero

$$\begin{aligned} E[Y(1)|D = 1] &= E[Y(1)|D = 0] = E[Y(1)] \\ \text{and } E[Y(0)|D = 0] &= E[Y(0)|D = 1] = E[Y(0)] \\ &\implies E[Y(0)|D = 1] - E[Y(0)|D = 0] = 0 \\ &\implies \text{Selection bias is 0} \\ &\implies \text{Naïve comparison} = \text{ATE on treated} \end{aligned}$$

The ideal experiment

A useful first question for any causal claim: **what experiment would you run if you could?**

- ▶ Who is randomized, into what treatment, and what outcome do you measure?
- ▶ If you can't describe the ideal experiment, the causal question may not be well-defined

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Marijuana and longevity:

- ▶ Randomly assign a group of people to smoke or not smoke marijuana for decades; compare age at death
- ▶ Randomization ensures smokers and non-smokers have the same $E[Y(0)] \implies$ no selection bias

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Often the ideal experiment is unethical, infeasible, or too expensive

- ▶ Rest of the course: approximating the ideal experiment with observational data

Randomization and regression

Assume treatment effects are constant and the same for everyone

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 &= \underbrace{\mathbf{E}[Y_i(0)]}_{\alpha} + \underbrace{(Y_i(1) - Y_i(0))}_{\beta} D_i + \underbrace{Y_i(0) - \mathbf{E}[Y_i(0)]}_{\eta_i \text{ (Random part of } Y_i(0))}
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Evaluate this with the treatment on ($D = 1$) and off ($D = 0$):

$$D = 1 \implies \mathbf{E}[Y|D = 1] = \alpha + \beta + \mathbf{E}[\eta|D = 1]$$

$$D = 0 \implies \mathbf{E}[Y|D = 0] = \alpha + \mathbf{E}[\eta|D = 0]$$

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Then:

$$\begin{aligned}
 \mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0] &= (\alpha + \beta + \mathbf{E}[\eta|D = 1]) - (\alpha + \mathbf{E}[\eta|D = 0]) \\
 &= \beta + \underbrace{(\mathbf{E}[\eta|D = 1] - \mathbf{E}[\eta|D = 0])}_{\text{selection bias}}
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 \end{aligned}$$

Selection bias will equal zero if the correlation between the error term and treatment assignment is zero.

- ▶ If D is randomly assigned, regression of Y onto D (and a constant) yields the causal effect β

Covariates and pre-treatment heterogeneity

Suppose in addition to outcome (Y) and treatment (D), there are **pre-treatment** or **pre-determined** covariates W

- ▶ e.g. interested in treatment effect across age

Treatment effect and naive comparison likely will not match will not match if there is pre-treatment heterogeneity

Assume: treatment assignment independent of covariates:

$$D \perp (Y(0), Y(1), W), \quad \text{and } 0 \leq P(D = 1) \leq 1$$

then:

$$\mathbf{E}[Y|D = d, W] = \mathbf{E}[Y(d)|D = d, W] = \mathbf{E}[Y(d)|W]$$

⇒ **conditional average treatment effects** can be estimated from conditional average predictive effects:

$$\mathbf{E}[Y(1)|W] - \mathbf{E}[Y(0)|W] = \mathbf{E}[Y|D = 1, W] - \mathbf{E}[Y|D = 0, W]$$

Bad controls

TODO

Sometimes tempting to control for post-treatment covariates

Example: Interested in the effect of college on earnings

- ▶ Suppose people can work white-collar or blue-collar job.

Good controls: can be thought of as having been fixed at the time the regressor of interest was determined.

Testing covariate balance

Random assignment assumption induces covariate balance.

- Distribution of covariates should be the same under treatment and control:

$$W|D = 1 \sim W|D = 0$$

equivalently:

$$D|W \sim D$$

Useful implication is:

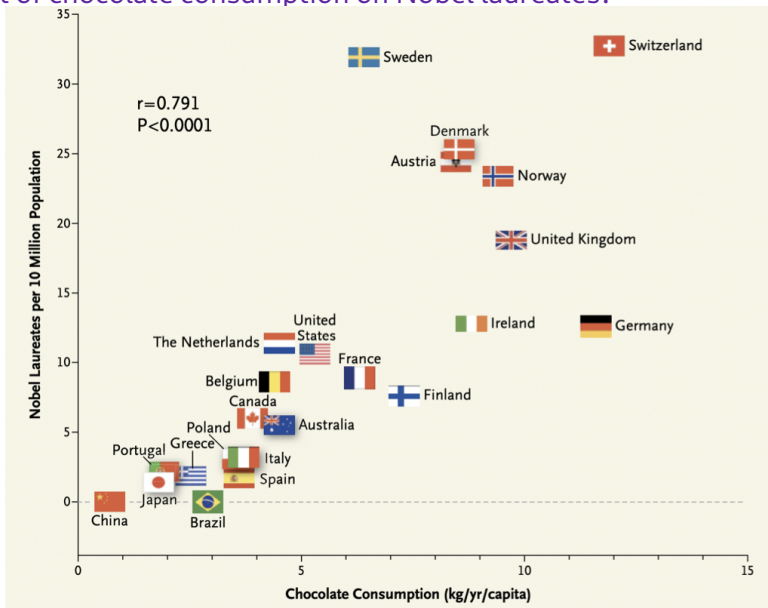
$$\mathbf{E}[D|W] = \mathbf{E}[D]$$

This is testable using regression tools; the R^2 of a regression of D on W is 0 under randomization.

Potential Outcomes

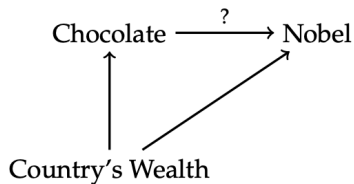
Conditional Independence Assumption

Impact of chocolate consumption on Nobel laureates?



Common cause

Common cause creates non-causal association



- Should be comparing Switzerland to a country similar to Switzerland

Conditional Independence Assumption

Last section: how do we infer causal effects when we have **random assignment** of treatment.

- ⇒ Independence of potential outcomes and the treatment
- ⇒ $Y(0), Y(1) \perp D$
- ⇒ randomization

What happens when we don't have independence? Meaning the occurrence of D influences the occurrence of $Y(0)$ and $Y(1)$ (or vice versa)?

Perhaps we can condition on certain variables X , such that the treatment D_i and the potential outcomes $Y_i(1), Y_i(0)$ LOOK independent

$$\{Y_i(0), Y_i(1)\} \perp D_i | X_i$$

- ▶ Think of treatment D_i as generated randomly (independent of potential outcomes) **conditional** on some pre-treatment variable X
- ⇒ can learn about average causal treatment effects by regression of Y_i on D_i and X_i .

Conditional independence

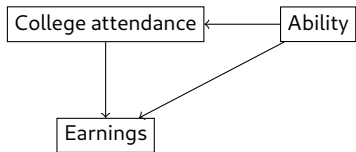
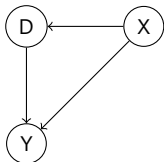
Conditional independence assumption:

$$\{Y_i(0), Y_i(1)\} \perp D_i | X_i$$

Intuitively: occurrence of D_i does not influence the occurrence of $Y_i(0)$ or $Y_i(1)$, or vice versa, conditional on knowing X_i :

$$\mathbf{E}[Y_i(0)|D_i, X_i] = \mathbf{E}[Y_i(0)|X_i]$$

$$\mathbf{E}[Y_i(1)|D_i, X_i] = \mathbf{E}[Y_i(1)|X_i]$$



Selection bias under CIA

Under conditional independence assumption, selection bias disappears conditional on X_i :

$$\begin{aligned} \mathbf{E}[Y_i(0)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] &= \mathbf{E}[Y_i(0)|X_i, D_i = 0] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] \\ &= 0 \end{aligned}$$

So we can write the difference in observed outcomes **conditional on X_i** as:

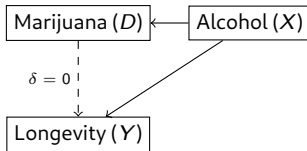
$$\begin{aligned} \mathbf{E}[Y_i|X_i, D_i = 1] - \mathbf{E}[Y_i|X_i, D_i = 0] &= \mathbf{E}[Y_i(1)|X_i, D_i = 1] - \mathbf{E}[Y_i(0)|X_i, D_i = 0] \\ &= \mathbf{E}[Y_i(1)|X_i] - \mathbf{E}[Y_i(0)|X_i] \\ &= \mathbf{E}[Y_i(1) - Y_i(0)|X_i] \end{aligned}$$

Given the CIA, comparisons of average earnings (conditional on X_i) have a causal interpretation

Example

Suppose you are studying the impact of marijuana on longevity.

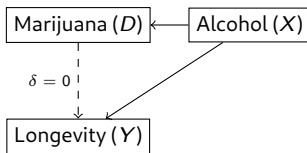
- ▶ As before, assume no causal effect:
 $Y = Y(0) = Y(1)$, so
 $\delta = \mathbf{E}[Y(1)] - \mathbf{E}[Y(0)] = 0$
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Example

Suppose you are studying the impact of marijuana on longevity.

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Without adjusting for X : higher D predicts higher X , which predicts lower Y

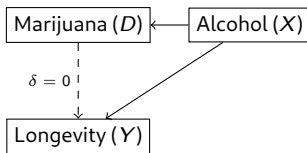
$$E[Y|D = 1] - E[Y|D = 0] < 0$$

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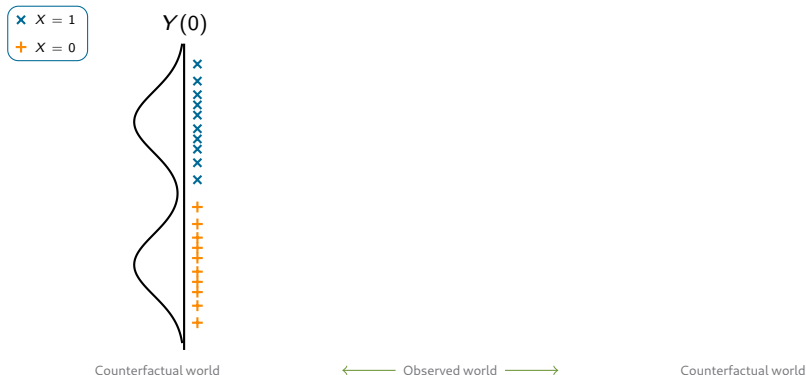
Conditioning on X removes the selection bias:

$$E[Y|D = 1, X] - E[Y|D = 0, X] = E[Y(1) - Y(0)|X]$$

provided that, conditional on X , variation in D is independent of potential outcomes:

- ▶ **Conditional ignorability:** $D \perp Y(d)|X$ for each d (i.e., the CIA)
- ▶ **Consistency:** $Y = Y(D)$

Selection on X creates selection bias



Potential outcome $Y(0)$ for everyone. Binary covariate X : units with $X = 1$ (x) tend to have higher outcomes than units with $X = 0$ (+)

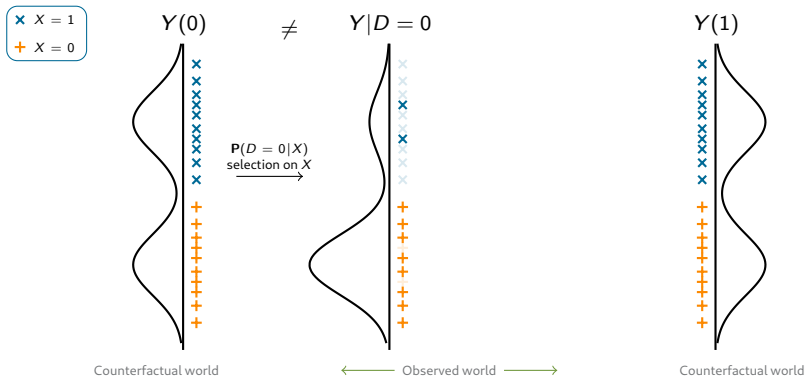
Source: adapted from Chernozhukov et al. (2025), Figure 5.4

Selection on X creates selection bias



No treatment effect: each unit's $Y(1)$ equals its $Y(0)$, so $Y(0)$ and $Y(1)$ have identical distributions ($\delta = 0$)

Selection on X creates selection bias

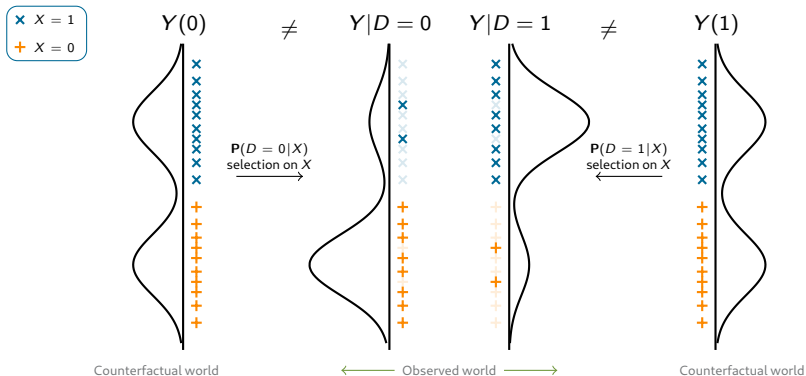


Selection on X : $X = 0$ units are more likely to be untreated, $P(D = 0|X = 0) > P(D = 0|X = 1)$

- We only observe $Y(0)$ for the untreated $\implies Y|D = 0$ is shifted down relative to $Y(0)$

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

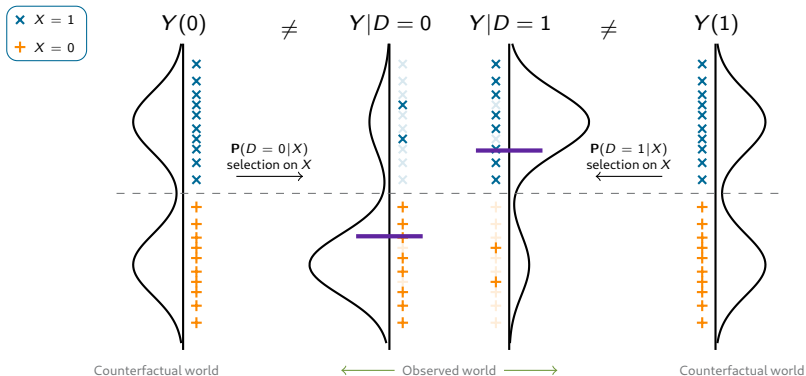
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$X = 1$ units are more likely to be treated, $P(D = 1|X = 1) > P(D = 1|X = 0)$

- ▶ We only observe $Y(1)$ for the treated $\implies Y|D = 1$ is shifted up relative to $Y(1)$

Selection on X creates selection bias



Naïve comparison: $E[Y|D = 1] > E[Y|D = 0]$, even though $E[Y(1)] = E[Y(0)]$ (dashed line)

- Difference is entirely **selection bias**: the treated and untreated have different mixes of X

Source: adapted from Chernozhukov et al. (2025), Figure 5.4

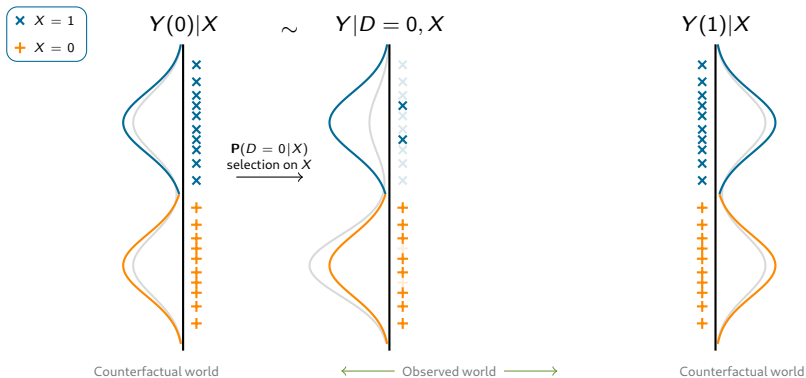
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Condition on X : split each potential outcome distribution by X

- $Y(0)|X = 1$ (blue) and $Y(0)|X = 0$ (orange); same for $Y(1)$

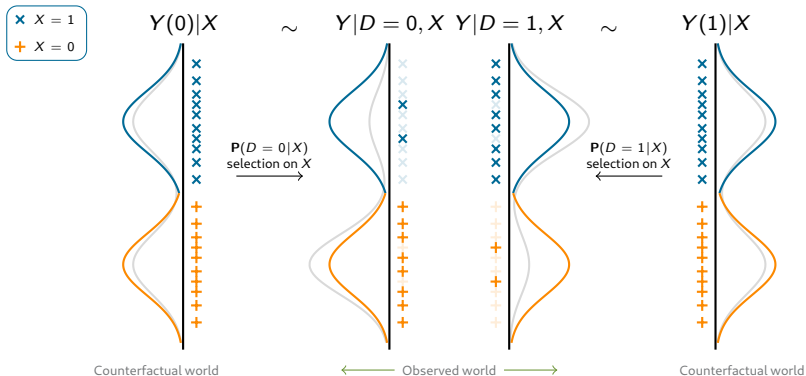
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Within $X = x$, selection is as good as random: untreated $X = 1$ units look like all $X = 1$ units

$\Rightarrow Y|D=0, X$ has the same distribution as $Y(0)|X$

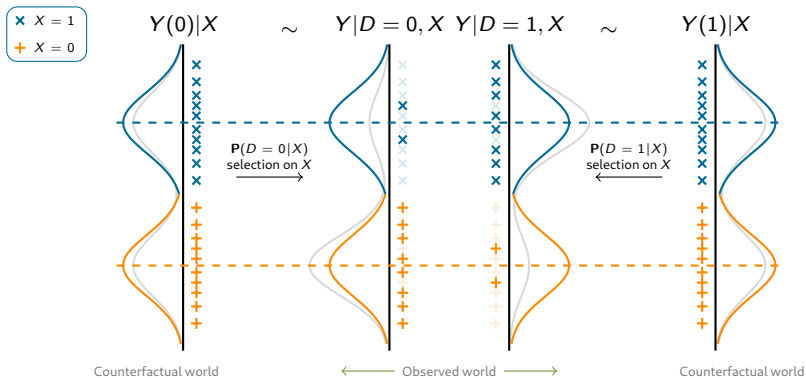
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Same for the treated: $Y|D=1, X$ has the same distribution as $Y(1)|X$

- Only the *mix* of X differs between treated and untreated, not outcomes within X

Conditioning on X removes selection bias



Within each $X = x$, observed means equal potential outcome means (dashed lines):

$$\begin{aligned}\mathbf{E}[Y|D = 1, X = x] - \mathbf{E}[Y|D = 0, X = x] &= \mathbf{E}[Y(1)|X = x] - \mathbf{E}[Y(0)|X = x] \\ &= \delta(x) = 0\end{aligned}$$

- Average $\delta(x)$ over the distribution of X to recover δ

Identification by conditioning

Ignorability says variation in D is **as good as random** conditional on X

- ▶ Among units with the same covariates ($X = x$), treatment variation $D|X = x$ is produced as *if* by an RCT
- ▶ “Ignorable”: distribution of $Y(d)$ depends only on X , not on D
- ▶ Also called **conditional exogeneity** or the **conditional independence assumption** (CIA) in econometrics

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For this to work, we need **both** treated and control units at every value of X

- ▶ The **propensity score** $p(X) := P(D = 1|X)$ plays a key role

Overlap assumption

Overlap (full support): the propensity score is non-degenerate

$$p(X) := P(D = 1|X), \quad P(0 < p(X) < 1) = 1$$

- ▶ Requires variation in D at each value x in the support of X
- ▶ Equivalently: $\text{support}(D, X) = \{0, 1\} \times \text{support}(X)$
- ▶ Conditional analogue of **positivity** under random assignment ($0 < P(D = 1) < 1$)

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Without overlap: at some values x there are only treated (or only control) units

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Example: if every heavy drinker smokes marijuana, then $p(\text{heavy drinker}) = 1$

- ▶ No non-smoking heavy drinkers to compare against

Conditioning on X removes selection bias

Assumptions:

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Theorem (conditioning on X removes selection bias): Under conditional ignorability and overlap,

$$\mathbf{E}[Y|D = d, X] = \mathbf{E}[Y(d)|D = d, X] = \mathbf{E}[Y(d)|X]$$

- The CEF of observed Y given $D = d$ and X recovers the CEF of the potential outcome $Y(d)$ given X

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Why?

- ▶ Overlap makes it possible to condition on $\{D = 0, X\}$ and $\{D = 1, X\}$ at every value in the support of X
- ▶ First equality: consistency
- ▶ Second equality: ignorability

Predictive effects and treatment effects

Predictive effects (computable from observed data):

Conditional avg. predictive effect (CAPE): $\pi(X) = \mathbf{E}[Y|D = 1, X] - \mathbf{E}[Y|D = 0, X]$

Avg. predictive effect (APE): $\pi = \mathbf{E}[\pi(X)]$

Treatment effects (defined by potential outcomes):

Conditional avg. treatment effect (CATE): $\delta(X) = \mathbf{E}[Y(1)|X] - \mathbf{E}[Y(0)|X]$

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By the theorem, the conditional average predictive effect equals the conditional average treatment effect:

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Averaging over X , the APE and ATE also agree:

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$$\delta = \mathbf{E}[\delta(X)] = \mathbf{E}[\pi(X)] = \pi$$

- ▶ $\pi(X)$ is computable from data $\implies$ we can learn $\delta(X)$ and δ
- ▶ Note: $\pi = \mathbf{E}[\pi(X)]$ generally differs from the naïve comparison $\mathbf{E}[Y|D = 1] - \mathbf{E}[Y|D = 0]$

When treatment takes on more than two values

Expanding CIA assumption to relations that can take on more than two values, such as when $s_i = \{0, 1, \dots, 16\}$ equals years of schooling:

$$Y_i(s) = f_i(s)$$

Individual specific function:

- ▶ Potential earnings that person i would receive after obtaining s years of education
- ▶ What would i learn for any value of schooling

If s could only take on two values, $s = 12$ and $s = 16$, then we're considering the value of college vs. no college:

$$Y_i(0) = f_i(12), \quad Y_i(1) = f_i(16)$$

CIA with non-binary treatment

CIA in this more general set-up:

$$Y_i(s) \perp s_i | X_i, \quad \forall s_i$$

- ▶ In randomized experiments (RCTs), s_i is randomly assigned based on covariates X_i
- ▶ In observational studies, the CIA means s_i is **as good as randomly assigned** conditional on X_i

Under conditional independence assumption, comparisons of average earnings across schooling levels have a causal interpretation:

$$\mathbf{E}[Y_i | X_i, s_i = s] - \mathbf{E}[Y_i | X_i, s_i = s - 1] = \mathbf{E}[f_i(s) - f_i(s - 1) | X_i]$$

For example, average causal effect of high school graduation:

$$\begin{aligned} \mathbf{E}[Y_i | X_i, s_i = 12] - \mathbf{E}[Y_i | X_i, s_i = 11] &= \mathbf{E}[f_i(12) | X_i, s_i = 12] - \mathbf{E}[f_i(11) | X_i, s_i = 11] \\ &= \mathbf{E}[f_i(12) - f_i(11) | X_i, s_i = 12] \end{aligned}$$

- ▶ Here, selection bias would come from the differences in potential earnings of high school graduates and non-graduates
- ▶ Under CIA, high school graduation is independent of potential earnings conditional on X_i , so selection bias disappears

CIA and regression

Regression is an empirical strategy that turns CIA into causal effects:

- ▶ Assume $f_i(s)$ is linear in s , and the same for everyone, except for an additive term η_i :

$$f_i(s) = \alpha + \rho s + \eta_i$$

- ▶ **Linear constant effects model:** equation is linear, and the functional relationship is the same for everyone
- ▶ Only individual-specific part of $f_i(s)$ is mean-zero η_i , which captures unobserved factors that determine potential earnings

Observed value will be:

$$Y_i = \alpha + \rho s_i + \eta_i,$$

Note that s_i is correlated with η_i : $\mathbf{E}[\eta_i | s_i] \neq 0$. If we ran this regression, we'd expect selection bias to impact our results.

CIA and regression (cont.)

Assume CIA holds, and η_i has the form:

$$\eta_i = X_i' \gamma + v_i$$

where:

- ▶ γ is a vector of population regression coefficients that satisfies $\mathbf{E}[\eta_i | X_i] = X_i' \gamma$
- ▶ v_i and X_i are uncorrelated by construction

Then, because of the CIA:

$$\begin{aligned} \mathbf{E}[f_i(s) | X_i, s_i] &= \mathbf{E}[f_i(s) | X_i] \\ &= \alpha + \rho s + \mathbf{E}[\eta_i | X] \\ &= \alpha + \rho s + X_i' \gamma \end{aligned}$$

The residual in the linear causal model v_i is uncorrelated with s_i and X_i , and ρ is the causal effect of interest:

$$Y_i = \alpha + \rho s_i + X_i' \gamma + v_i$$

Regression takeaways

We just discussed how CIA + linear causal model yields a causal interpretation of CEF

- ▶ Linear causal model: $Y_i = \alpha + \rho s_i + \eta_i$
- ▶ If CIA holds given some variables X_i : $Y_{0i}, Y_{1i} \perp s_i | X_i$
- ▶ Our regression function has a causal interpretation: $Y_i = \alpha + \rho s_i + X_i' \gamma + v_i$
 - ρ is the causal effect of s_i on Y_i
- ▶ CEF given by:

$$\mathbf{E}[Y_i | X_i, s_i] = \alpha + \rho s_i + X_i' \gamma$$

Thing to note:

- ▶ CIA is the same as unconfoundedness is the same as selection on observables
- ▶ In regression, CIA is the same as exogeneity